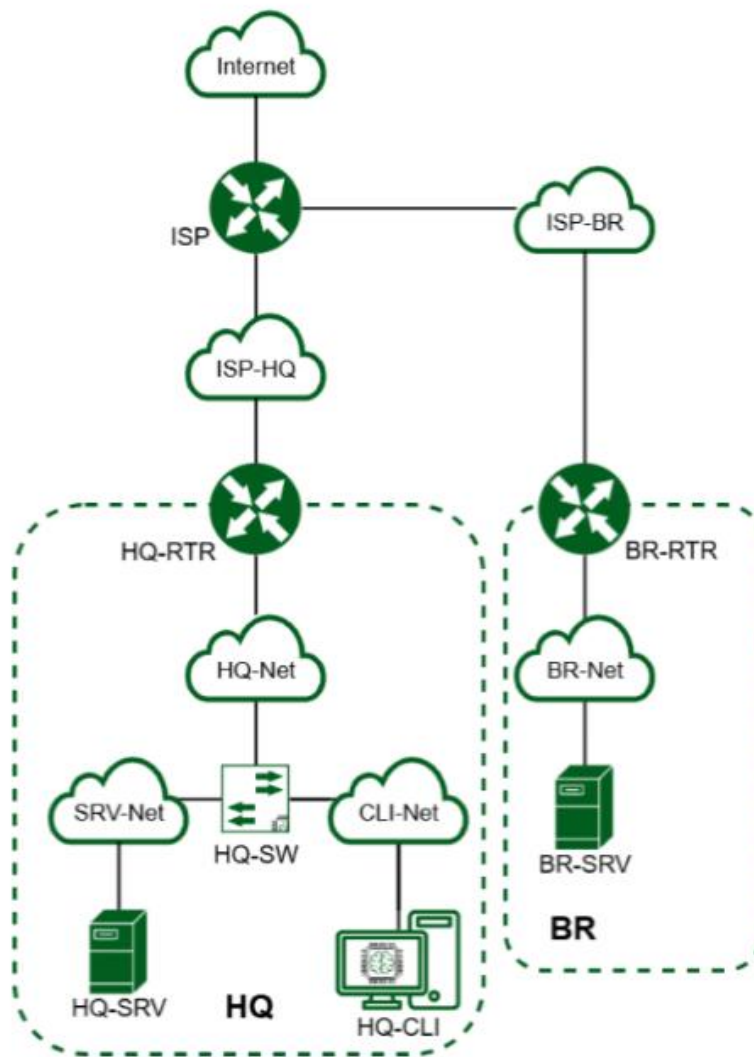


Топология сети:



Используемые ОС:

ISP, HQ-RTR, BR-RTR - Astra Linux CE

HQ-SRV, BR-SRV - Альт Сервер 10

HQ-CLI - Альт Рабочая станция 10

Таблица адресации:

Устройство	Интерфейс	IP адрес	Маска подсети	Шлюз по умолчанию
ISP	eth0	DHCP (автоматически)	-	DHCP (автоматически)
ISP	eth1	172.16.1.1/28	255.255.255.240	-
ISP	eth2	172.16.2.1/28	255.255.255.240	-
HQ-RTR	eth0	172.16.1.2/28	255.255.255.240	172.16.1.1
HQ-RTR	eth1.100	192.168.1.1/27	255.255.255.224	-
HQ-RTR	eth1.200	192.168.2.1/28	255.255.255.240	-
HQ-RTR	eth1.999	192.168.99.1/29	255.255.255.248	-
HQ-RTR	gre1 (ip tunnel)	10.0.0.1/30	255.255.255.252	-
HQ-SRV	ens18	192.168.1.2/27	255.255.255.224	192.168.1.1
HQ-CLI	ens18	DHCP (192.168.2.2/28)	255.255.255.240	DHCP (192.168.2.1)
BR-RTR	eth0	172.16.2.2/28	255.255.255.240	172.16.2.1
BR-RTR	eth1	192.168.3.1/28	255.255.255.240	-
BR-SRV	ens18	192.168.3.2/28	255.255.255.240	192.168.3.1

- команды

- проверка

Модуль 1. Настройка сетевой инфраструктуры

ISP (user P@ssw0rd):

Задание 1. Произведите базовую настройку устройств

sudo su

hostnamectl set-hostname isp.au-team.irpo; exec bash

Задание 11. Настройте часовой пояс на всех устройствах

timedatectl set-timezone Asia/Yekaterinburg

timedatectl

Задание 2. Настройте доступ к сети Интернет, на маршрутизаторе ISP

mcedit /etc/network/interfaces

```
/etc/network/interfaces [-M--] 0 L:[ 1+20 21/ 21] *(387 / 387b) <EOF>
# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback

auto eth0
iface eth0 inet dhcp

auto eth1
iface eth1 inet static
<----->address 172.16.1.1/28

auto eth2
iface eth2 inet static
<----->address 172.16.2.1/28
```

echo "net.ipv4.ip_forward=1" >> /etc/sysctl.conf

sysctl -p

iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE

iptables-save > /etc/rules.v4

crontab -e

в конец добавить:

@reboot /sbin/iptables-restore < /etc/rules.v4

@reboot /sbin/sysctl -p

echo "nameserver 8.8.8.8" > /etc/resolv.conf

systemctl restart networking

ip -br -c a

ping 8.8.8.8

HQ-RTR (user P@ssw0rd):

Задание 1. Произведите базовую настройку устройств + туннель

```
sudo su
hostnamectl set-hostname hq-rtr.au-team.irpo; exec bash
mcedit /etc/network/interfaces
```

```
GNU nano 2.7.4                                     Файл: /etc/network/interfaces
# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback

auto eth0
iface eth0 inet static
    address 172.16.1.2/28
    gateway 172.16.1.1

auto eth1
iface eth1 inet manual

auto eth1.100
iface eth1.100 inet static
    address 192.168.1.1/27
    vlan-raw-device eth1

auto eth1.200
iface eth1.200 inet static
    address 192.168.2.1/28
    vlan-raw-device eth1

auto eth1.999
iface eth1.999 inet static
    address 192.168.99.1/29
    vlan-raw-device eth1

auto gre1
iface gre1 inet tunnel
    address 10.0.0.1
    netmask 255.255.255.252
    mode gre
    local 172.16.1.2
    endpoint 172.16.2.2
    ttl 255
```

```
systemctl restart networking
```

```
ip -br -c a
ping 8.8.8.8
```

Задание 11. Настройте часовой пояс на всех устройствах

```
timedatectl set-timezone Asia/Yekaterinburg
timedatectl
```

Задание 3. Создайте локальные учетные записи на серверах HQ-SRV и BR-SRV:

```
useradd -m net_admin -u 2026
passwd net_admin
usermod -aG sudo net_admin
nano /etc/sudoers
```

```
root ALL=(ALL:ALL) ALL

# Allow members of group sudo to execute any command
%sudo ALL=(ALL:ALL) NOPASSWD: ALL

# See sudoers(5) for more information on "#include"
```

Проверка:

```
root@hq-rtr:~# su net_admin
sh-4.4$ sudo ls
sudo: unable to resolve host hq-rtr.au-team.irpo: В соединении отказано
Desktop Desktops needs SystemWallpapers Вугео Документы Загрузки Изображения Музыка Общедоступные Шаблоны
sh-4.4$ su root
Пароль:
root@hq-rtr:~#
```

Задание 8. Настройка динамической трансляции адресов маршрутизаторах HQ-RTR и BR-RTR

```
echo "net.ipv4.ip_forward=1" >> /etc/sysctl.conf
sysctl -p
iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE
iptables-save > /etc/rules.v4
echo "nameserver 8.8.8.8" > /etc/resolv.conf
crontab -e
# в конец добавить:
@reboot /sbin/iptables-restore < /etc/rules.v4
@reboot /sbin/sysctl -p
systemctl restart networking
```

Задание 7. Обеспечьте динамическую маршрутизацию на маршрутизаторах HQ-RTR и BR-RTR: сети одного офиса должны быть доступны из другого офиса и наоборот

```
echo "deb [trusted=yes] https://archive.debian.org/debian buster main" >>
/etc/apt/sources.list
apt-get update && apt-get install frr
mcedit /etc/frr/daemons → ospfd=yes
systemctl restart frr
systemctl enable frr
vtysh

conf t
router ospf
network 10.0.0.0/30 area 0
network 192.168.1.0/27 area 0
network 192.168.2.0/28 area 0
network 192.168.99.0/29 area 0
```

```

exit
do wr mem
int gre1
ip ospf authentication message-digest
ip ospf message-digest-key 1 md5 P@ssw0rd
quit
do wr mem
show run
show ip ospf route
exit
systemctl restart frr
systemctl enable frr
crontab -e

```

В КОНЕЦ ДОБАВИТЬ:

```

*/2 * * * * /bin/systemctl restart frr
@reboot /sbin/iptables-restore < /etc/rules.v4
@reboot /sbin/sysctl -p

```

Задание 9. Настройте протокол динамической конфигурации хостов для сети в сторону HQ-CLI

```

apt-get install -y isc-dhcp-server
nano /etc/default/isc-dhcp-server

```

```

# Defaults for isc-dhcp-server (sourced by /etc/init.d/isc-dhcp-server)

# Path to dhcpd's config file (default: /etc/dhcp/dhcpd.conf).
#DHCPDv4_CONF=/etc/dhcp/dhcpd.conf
#DHCPDv6_CONF=/etc/dhcp/dhcpd6.conf

# Path to dhcpd's PID file (default: /var/run/dhcpd.pid).
#DHCPDv4_PID=/var/run/dhcpd.pid
#DHCPDv6_PID=/var/run/dhcpd6.pid

# Additional options to start dhcpd with.
# Don't use options -cf or -pf here; use DHCPD_CONF/ DHCPD_PID instead
#OPTIONS=""

# On what interfaces should the DHCP server (dhcpd) serve DHCP requests?
# Separate multiple interfaces with spaces, e.g. "eth0 eth1".
INTERFACESv4="eth1.200"
#INTERFACESv6=""

```

```

nano /etc/dhcp/dhcpd.conf

```

```
GNU nano 2.7.4                               Файл: /etc/dhcp/dhcpd.conf
# dhcpd.conf
#
# Sample configuration file for ISC dhcpd
#
# option definitions common to all supported networks
#option domain-name "example.org";
#option domain-name-servers ns1.example.org, ns2.example.org;

default-lease-time 600;
max-lease-time 7200;

# The ddns-updates-style parameter controls whether or not the server will
# attempt to do a DNS update when a lease is confirmed. We default to the
# behavior of the version 2 packages ('none', since DHCP v2 didn't
# have support for DDNS.)
ddns-update-style none;

# If this DHCP server is the official DHCP server for the local
# network, the authoritative directive should be uncommented.
authoritative;
```

А также в конце пишем:

```
# A slightly different configuration for an internal subnet.
subnet 192.168.2.0 netmask 255.255.255.240 {
    range 192.168.2.2 192.168.2.10;
    option domain-name-servers 192.168.1.2;

    option routers 192.168.2.1;
}
```

systemctl restart isc-dhcp-server

BR-RTR: (user P@ssw0rd):

Задание 1. Произведите базовую настройку устройств + туннель

```
sudo su
hostnamectl set-hostname br-rtr.au-team.irpo; exec bash
mcedit /etc/network/interfaces
```

```
/etc/network/interfaces [----] 21 L:[ 1+24 25/ 27] *(522 / 537b) 0046 0x02E
# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback

auto eth0
iface eth0 inet static
    address 172.16.2.2/28
    gateway 172.16.2.1

auto eth1
iface eth1 inet static
    address 192.168.3.1/28

auto gre1
iface gre1 inet tunnel
    address 10.0.0.2
    netmask 255.255.255.252
    mode gre
    local 172.16.2.2
    endpoint 172.16.1.2
    ttl 255
```

```
systemctl restart networking
```

```
ip -br -c a
ping 8.8.8.8
```

Задание 11. Настройте часовой пояс на всех устройствах

```
timedatectl set-timezone Asia/Yekaterinburg
timedatectl
```

Задание 3. Создайте локальные учетные записи на серверах HQ-SRV и BR-SRV:

```
useradd -m net_admin -u 2026
passwd net_admin
usermod -aG sudo net_admin
nano /etc/sudoers
```

```
root ALL=(ALL:ALL) ALL

# Allow members of group sudo to execute any command
%sudo ALL=(ALL:ALL) NOPASSWD: ALL

# See sudoers(5) for more information on "#include"
```


Проверка:

```
root@hq-rtr:~# su net_admin
sh-4.4$ sudo ls
sudo: unable to resolve host hq-rtr.au-team.irpo: В соединении отказано
Desktop Desktops needs SystemWallpapers Вугео Документы Загрузки Изображения Музыка Общедоступные Шаблоны
sh-4.4$ su root
Пароль:
root@hq-rtr:~#
```

Задание 8. Настройка динамической трансляции адресов маршрутизаторах HQ-RTR и BR-RTR

```
echo "net.ipv4.ip_forward=1" >> /etc/sysctl.conf
sysctl -p
iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE
iptables-save > /etc/rules.v4
echo "nameserver 8.8.8.8" > /etc/resolv.conf
crontab -e
# в конец добавить:
@reboot /sbin/iptables-restore < /etc/rules.v4
@reboot /sbin/sysctl -p
systemctl restart networking
```

Задание 7. Обеспечьте динамическую маршрутизацию на маршрутизаторах HQ-RTR и BR-RTR: сети одного офиса должны быть доступны из другого офиса и наоборот

```
echo "deb [trusted=yes] https://archive.debian.org/debian buster main" >>
/etc/apt/sources.list
apt-get update && apt-get install frr
mcedit /etc/frr/daemons → ospfd=yes
systemctl restart frr
systemctl enable frr
vtysh

conf t
router ospf
network 10.0.0.0/30 area 0
network 192.168.3.0/28 area 0
exit
do wr mem
int gre1
ip ospf authentication message-digest
ip ospf message-digest-key 1 md5 P@ssw0rd
quit
do wr mem
show run
show ip ospf route
exit
systemctl restart frr
```

```
systemctl enable frr  
crontab -e
```

В КОНЕЦ ДОБАВИТЬ:

```
*/2 * * * * /bin/systemctl restart frr  
@reboot /sbin/iptables-restore < /etc/rules.v4  
@reboot /sbin/sysctl -p
```

Проверка сервиса:

Возвращаемся на HQ-RTR

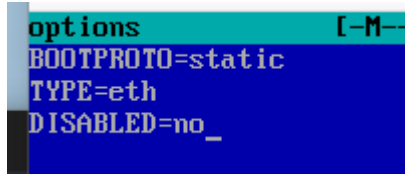
```
vtysh  
conf t  
show ip ospf neighbor  
show ip ospf route
```

```
hq-rtr.au-team.irpo(config-router)# do show ip ospf route  
===== OSPF network routing table =====  
N    10.0.0.0/30          [10] area: 0.0.0.0  
      directly attached to gre1  
N    192.168.1.0/27      [1] area: 0.0.0.0  
      directly attached to eth1.100  
N    192.168.2.0/28      [1] area: 0.0.0.0  
      directly attached to eth1.200  
N    192.168.3.0/28      [11] area: 0.0.0.0  
      via 10.0.0.2, gre1  
N    192.168.99.0/29     [1] area: 0.0.0.0  
      directly attached to eth1.999  
  
===== OSPF router routing table =====  
  
===== OSPF external routing table =====  
  
hq-rtr.au-team.irpo(config-router)#
```

HQ-SRV(root P@ssw0rd):

Задание 1. Произведите базовую настройку устройств

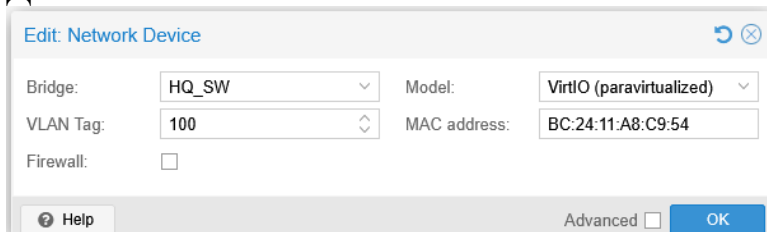
```
hostnamectl set-hostname hq-srv.au-team.irpo; exec bash  
mcedit /etc/net/ifaces/ens18/options
```



```
options  
BOOTPROTO=static  
TYPE=eth  
DISABLED=no_
```

```
echo "192.168.1.2/27" > /etc/net/ifaces/ens18/ipv4address  
echo "default via 192.168.1.1" > /etc/net/ifaces/ens18/ipv4route  
echo "nameserver 8.8.8.8" > /etc/resolv.conf  
systemctl restart networking  
ip -br -c a  
ping 8.8.8.8
```

Добавим VLAN-тег:



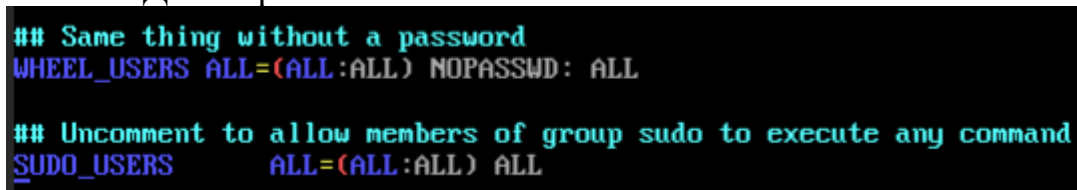
Задание 11. Настройте часовой пояс на всех устройствах

```
timedatectl set-timezone Asia/Yekaterinburg  
timedatectl
```

Задание 3. Создайте локальные учетные записи на серверах HQ-SRV и BR-SRV:

```
mcedit /etc/sudoers
```

Меняем ДВЕ строки:



```
## Same thing without a password  
WHEEL_USERS ALL=(ALL:ALL) NOPASSWD: ALL  
  
## Uncomment to allow members of group sudo to execute any command  
SUDO_USERS ALL=(ALL:ALL) ALL
```

```
sudo useradd sshuser -u 2026  
passwd sshuser  
sudo usermod -aG wheel sshuser
```

Проверка:

```

[root@br-srv ~]# su sshuser
bash: /root/.bashrc: Permission denied
bash-4.4$ sudo ls
tmp
bash-4.4$ su root
Password:
su: Authentication failure
bash-4.4$ su root
Password:
[root@br-srv ~]#

```

Задание 5. Настройте безопасный удаленный доступ на серверах HQ-SRV и BR-SRV

mcedit /etc/openssh/sshd_config

```

Port 2026
AllowUsers sshuser
PermitRootLogin no
MaxAuthTries 2
Banner /root/banner_

```

mcedit /root/banner

```

banner [-M--]
Authorized access only
-

```

systemctl restart sshd

systemctl enable sshd

Проверка:

```

[root@hq-srv ~]# ssh user@localhost -p 2026
Authorized access only
user@localhost's password:
ssh: Permission denied, please try again.
user@localhost's password:

[root@hq-srv ~]# ssh sshuser@localhost -p 2026
Authorized access only
sshuser@localhost's password:
[sshuser@hq-srv ~]#

```

Задание 10. Настройте инфраструктуру разрешения доменных имён для офисов HQ и BR

apt-get update && apt-get install dnsmasq -y

mcedit /etc/dnsmasq.conf

```

interface=*
server=8.8.8.8
#no-resolv
#no-hosts

host-record=hq-rtr.au-team.irpo,192.168.1.1
host-record=hq-srv.au-team.irpo,192.168.1.2
host-record=hq-cli.au-team.irpo,192.168.2.2

address=/br-rtr.au-team.irpo/192.168.3.1
address=/br-srv.au-team.irpo/192.168.3.2
address=/docker.au-team.irpo/172.16.1.1
address=/web.au-team.irpo/172.16.2.1

```

systemctl restart dnsmasq

systemctl enable --now dnsmasq

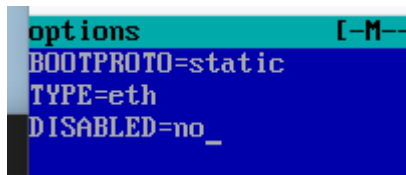
Проверка:

```
[root@hq-srv ~]# ping hq-rtr.au-team.irpo
PING hq-rtr.au-team.irpo (192.168.1.1) 56(84) bytes of data.
64 bytes from _gateway (192.168.1.1): icmp_seq=1 ttl=64 time=1.17 ms
64 bytes from _gateway (192.168.1.1): icmp_seq=2 ttl=64 time=1.10 ms
^C
--- hq-rtr.au-team.irpo ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 999ms
rtt min/avg/max/mdev = 1.095/1.134/1.174/0.039 ms
[root@hq-srv ~]#
```

BR-SRV(root P@ssw0rd):

Задание 1. Произведите базовую настройку устройств

```
hostnamectl set-hostname br-srv.au-team.irpo; exec bash
mcedit /etc/net/ifaces/ens18/options
```



```
options
BOOTPROTO=static
TYPE=eth
DISABLED=no_
```

```
echo "192.168.3.2/28" > /etc/net/ifaces/ens18/ipv4address
echo "default via 192.168.3.1" > /etc/net/ifaces/ens18/ipv4route
echo "nameserver 8.8.8.8" > /etc/resolv.conf
systemctl restart networking
ip -br -c a
ping 8.8.8.8
```

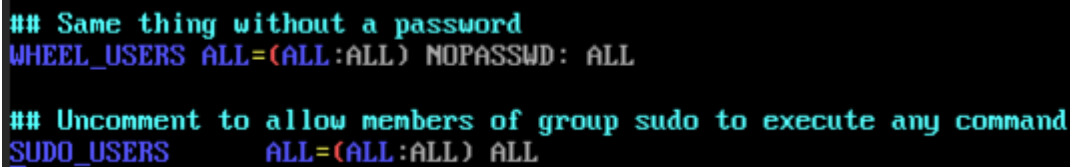
Задание 11. Настройте часовой пояс на всех устройствах

```
timedatectl set-timezone Asia/Yekaterinburg
timedatectl
```

Задание 3. Создайте локальные учетные записи на серверах HQ-SRV и BR-SRV:

```
mcedit /etc/sudoers
```

Меняем ДВЕ строки:

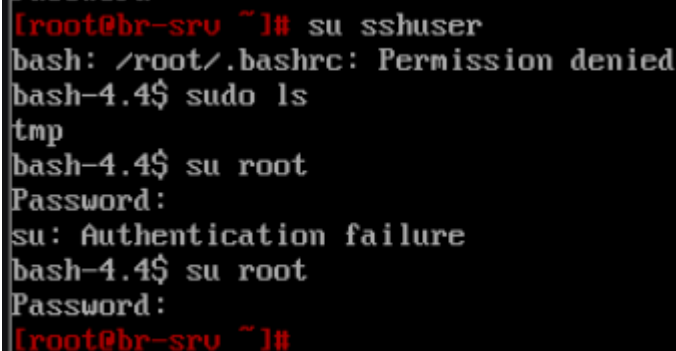


```
## Same thing without a password
WHEEL_USERS ALL=(ALL:ALL) NOPASSWD: ALL

## Uncomment to allow members of group sudo to execute any command
SUDO_USERS ALL=(ALL:ALL) ALL
```

```
sudo useradd sshuser -u 2026
passwd sshuser
sudo usermod -aG wheel sshuser
```

Проверка:



```
[root@br-srv ~]# su sshuser
bash: /root/.bashrc: Permission denied
bash-4.4$ sudo ls
tmp
bash-4.4$ su root
Password:
su: Authentication failure
bash-4.4$ su root
Password:
[root@br-srv ~]#
```

Задание 5. Настройте безопасный удаленный доступ на серверах HQ-SRV и BR-SRV

```
mcedit /etc/openssh/sshd_config
```

```
Port 2026
AllowUsers sshuser
PermitRootLogin no
MaxAuthTries 2
Banner /root/banner_
```

mcedit /root/banner

```
banner [-M--]
Authorized access only
_
```

systemctl restart sshd

systemctl enable sshd

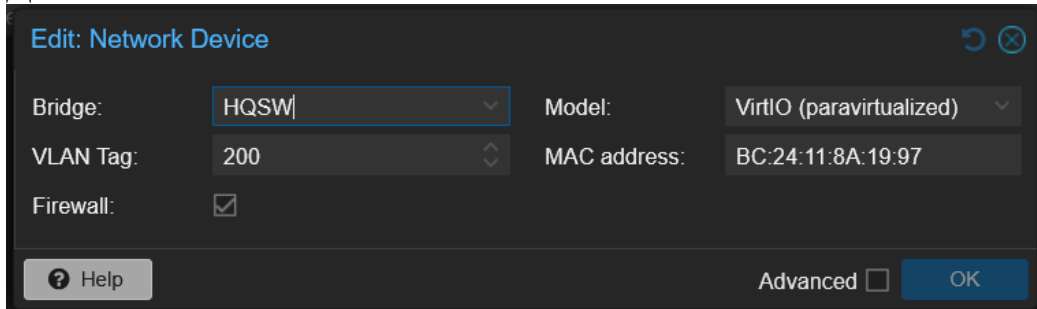
Проверка:

```
(root@hq-srv ~)# ssh user@localhost -p 2026
Authorized access only
user@localhost's password:
ssh: Permission denied, please try again.
user@localhost's password:

(root@hq-srv ~)# ssh sshuser@localhost -p 2026
Authorized access only
sshuser@localhost's password:
[sshuser@hq-srv ~]$
```

HQ-CLI(user P@ssw0rd):

Добавим VLAN-тег:



Edit: Network Device

Bridge: HQSWM Model: VirtIO (paravirtualized)

VLAN Tag: 200 MAC address: BC:24:11:8A:19:97

Firewall: ☒

Help Advanced ☐ OK

Открываем терминал.

su

Вводим пароль: P@ssw0rd

Задание 1. Произведите базовую настройку устройств

hostnamectl set-hostname hq-cli.au-team.irpo; exec bash

Задание 11. Настройте часовой пояс на всех устройствах

timedatectl set-timezone Asia/Yekaterinburg

timedatectl

Модуль 2. Настройка сетевой инфраструктуры

ISP (user P@ssw0rd):

Задание 4. Настройте службу сетевого времени на базе сервиса chrony на маршрутизаторе ISP:

```
apt-get update && apt-get install chrony -y
nano/etc/chrony/chrony.conf
```

Комментируем старый ntp:

```
#Welcome to the chrony configuration file. See chrony.conf(5) for more
# information about usable directives.
#pool 2.debian.pool.ntp.org iburst
```

И в конец файла пишем:

```
server 0.ru.pool.ntp.org iburst prefer minstratum 4
local stratum 5
allow 0.0.0.0/0
```

```
systemctl restart chronyd
```

Проверка:

```
root@isp:/home/igor# chronyc tracking
Reference ID      : 7F7F0101 ()
Stratum          : 5
Ref time (UTC)   : Sat Nov 22 09:37:37 2025
System time      : 0.000000000 seconds slow of NTP time
Last offset      : +0.000000000 seconds
RMS offset       : 0.000000000 seconds
Frequency        : 20.809 ppm slow
Residual freq    : +0.000 ppm
Skew             : 0.000 ppm
Root delay       : 0.000000000 seconds
Root dispersion  : 0.000000000 seconds
Update interval  : 0.0 seconds
Leap status      : Normal
root@isp:/home/igor# chronyc sources
210 Number of sources = 1
MS Name/IP address      Stratum Poll Reach LastRx Last sample
-----
^* mskm9-ntp02c.ntppool.yan> 4 6 17 0 -73us[-1393us] +/- 17ms
```

Задание 9. Настройте веб-сервер nginx как обратный прокси-сервер на ISP +

Задание 10. На маршрутизаторе ISP настройте web-based аутентификацию

```
apt-get install ssh nginx apache2-utils -y
htpasswd -bc /etc/nginx/.htpasswd WEB 'P@ssw0rd'
chmod 644 /etc/nginx/.htpasswd
rm -rf /etc/nginx/sites-available/default
mcedit /etc/nginx/sites-available/default
```

```

/etc/nginx/sites-available/default  [----]  0 L:[ 1+27 28/ 28] *(669 / 669b) <EOF>
server {
<----->listen 80;
<----->server_name web.au-team.irpo;
<----->
<----->location / {
<-----><----->proxy_pass http://172.16.1.2:8080;
<-----><----->proxy_set_header Host $host;
<-----><----->proxy_set_header X-Real-IP $remote_addr;
<-----><----->proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
<-----><----->proxy_set_header X-Forwarded-Proto $scheme;
<-----><----->auth_basic 'Web-Authorization';
<-----><----->auth_basic_user_file /etc/nginx/.htpasswd;
<----->}
}
server {
<----->listen 80;
<----->server_name docker.au-team.irpo;
<----->
<----->location / {
<-----><----->proxy_pass http://172.16.2.2:8080;
<-----><----->proxy_set_header Host $host;
<-----><----->proxy_set_header X-Real-IP $remote_addr;
<-----><----->proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
<-----><----->proxy_set_header X-Forwarded-Proto $scheme;
<----->}
}

```

nginx -t

systemctl restart nginx

HQ-RTR (user P@ssw0rd):

Задание 8. На маршрутизаторах сконфигурируйте статическую трансляцию портов

```
apt-get update && apt-get install ssh  
iptables -t nat -A PREROUTING -d 172.16.1.2 -p tcp --dport 8080 -j DNAT --to-  
destination 192.168.1.2:80  
iptables -t nat -A PREROUTING -d 172.16.1.2 -p tcp --dport 2026 -j DNAT --to-  
destination 192.168.1.2:2026  
iptables-save > /etc/rules.v4
```

BR-RTR: (user P@ssw0rd):

Задание 4. Настройте службу сетевого времени на базе сервиса chrony на маршрутизаторе ISP:

```
apt-get update && apt-get install ssh chrony -y  
mcedit /etc/chrony/chrony.conf
```

Комментируем следующие строки:

```
GNU nano 2.7.4                                Файл: /etc/chrony/chrony.conf  
  
# Welcome to the chrony configuration file. See chrony.conf(5) for more  
# information about usable directives.  
#pool 2.debian.pool.ntp.org iburst  
  
# This directive specify the location of the file containing ID/key pairs for  
# NTP authentication.  
keyfile /etc/chrony/chrony.keys  
  
# This directive specify the file into which chronyd will store the rate  
# information.  
driftfile /var/lib/chrony/chrony.drift  
  
# Uncomment the following line to turn logging on.  
#log tracking measurements statistics  
  
# Log files location.  
logdir /var/log/chrony  
  
# Stop bad estimates upsetting machine clock.  
maxupdateskew 100.0  
  
# This directive enables kernel synchronisation (every 11 minutes) of the  
# real-time clock. Note that it can't be used along with the 'rtcfile' directive.  
#rtcsync
```

В конце прописываем:

```
server 172.16.1.1 iburst
```

```
systemctl restart chrony  
systemctl enable chrony
```

Задание 8. На маршрутизаторах сконфигурируйте статическую трансляцию портов

```
apt-get update && apt-get install ssh  
iptables -t nat -A PREROUTING -d 172.16.2.2 -p tcp --dport 8080 -j DNAT --to-  
destination 192.168.3.2:8080  
iptables -t nat -A PREROUTING -d 172.16.2.2 -p tcp --dport 2026 -j DNAT --to-  
destination 192.168.3.2:2026  
iptables-save > /etc/rules.v4
```

HQ-SRV(root P@ssw0rd):

Задание 10. Настройте инфраструктуру разрешения доменных имён для офисов HQ и BR

```
mcedit /etc/dnsmasq.conf
```

Добавите строку над `server = 8.8.8.8`:

```
server=/au-team.irpo/192.168.3.2  
server=8.8.8.8
```

Задание 4. Настройте службу сетевого времени на базе сервиса chrony на маршрутизаторе ISP

```
apt-get update && apt-get install ssh chrony -y  
mcedit /etc/chrony/chrony.conf
```

Комментируем следующие строки:

```
GNU nano 2.7.4                                Файл: /etc/chrony/chrony.conf  
# Welcome to the chrony configuration file. See chrony.conf(5) for more  
# information about usable directives.  
#pool 2.debian.pool.ntp.org iburst  
  
# This directive specify the location of the file containing ID/key pairs for  
# NTP authentication.  
keyfile /etc/chrony/chrony.keys  
  
# This directive specify the file into which chronyd will store the rate  
# information.  
driftfile /var/lib/chrony/chrony.drift  
  
# Uncomment the following line to turn logging on.  
#log tracking measurements statistics  
  
# Log files location.  
logdir /var/log/chrony  
  
# Stop bad estimates upsetting machine clock.  
maxupdateskew 100.0  
  
# This directive enables kernel synchronisation (every 11 minutes) of the  
# real-time clock. Note that it can't be used along with the 'rtcfile' directive.  
#rtcsync
```

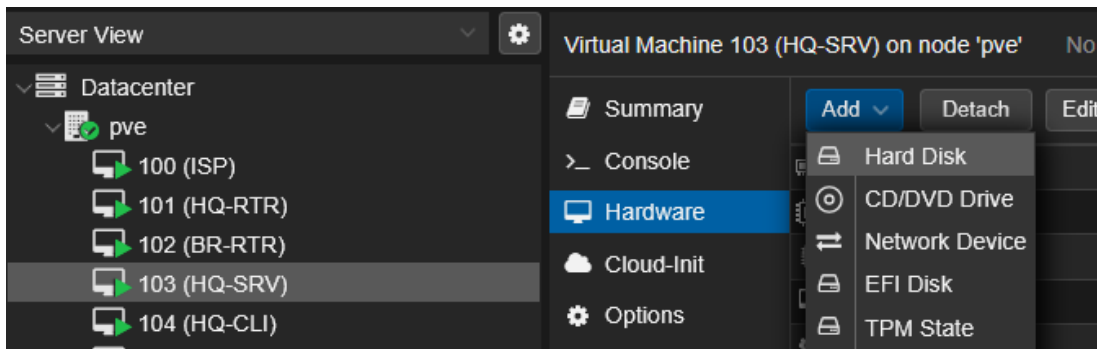
В конце прописываем:

```
server 172.16.1.1 iburst
```

```
systemctl restart chrony  
systemctl enable chrony
```

Задание 2. Сконфигурируйте файловое хранилище на сервере HQ-SRV

Для начала нужно подключить пару дисков по 1Гб к машине для дальнейшего их объединения в RAID 0.



lsblk

```
(root@hq-srv ~)# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda          8:0    0   32G  0 disk
├─sda1       8:1    0   977M  0 part [SWAP]
└─sda2       8:2    0   31G  0 part /
sdb          8:16   0    1G  0 disk
sdc          8:32   0    1G  0 disk
sr0         11:0    1   5.1G  0 rom
```

```
/sbin/mdadm --create /dev/md0 --level=0 --raid-device=2 /dev/sdb /dev/sdc
```

```
(root@hq-srv ~)# mdadm --create /dev/md0 --level=0 --raid-device=2 /dev/sdb /dev/sdc
mdadm: Defaulting to version 1.2 metadata
mdadm: array /dev/md0 started.
(root@hq-srv ~)#
```

```
/sbin/mdadm --detail --scan > /etc/mdadm.conf
```

```
(root@hq-srv ~)# cat /etc/mdadm.conf
ARRAY /dev/md0 metadata=1.2 name=hq-srv.au-team.irpo:0 UUID=b144da30:cf492519:a3f6651b:99b69c83
```

```
fdisk /dev/md0
```

```
(root@hq-srv ~)# fdisk /dev/md0

Welcome to fdisk (util-linux 2.39.2).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS (MBR) disklabel with disk identifier 0x1a7e26ab.

Command (m for help): _
```

1. n
2. p
- 3,4,5. skip
6. w

lsblk /dev/md0

```
(root@hq-srv ~)# lsblk /dev/md0
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
md0          9:0    0    2G  0 raid0
└─md0p1     259:0   0    2G  0 part
```

```
mkfs.ext4 /dev/md0p1
```

```
[root@hq-srv ~]# mkfs.ext4 /dev/md0p1
mke2fs 1.46.2 (28-Feb-2021)
Discarding device blocks: done
Creating filesystem with 523808 4k blocks and 130816 inodes
Filesystem UUID: 5e0e2b26-0d23-451d-9197-b2fcc49f80f8
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done
```

mkdir /raid

mount /dev/md0p1 /raid

```
[root@hq-srv ~]# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE  MOUNTPOINTS
sda          8:0    0   32G  0 disk
├─sda1       8:1    0   977M  0 part  [SWAP]
└─sda2       8:2    0   31G  0 part  /
sdb          8:16   0    1G  0 disk
└─md0        9:0    0    2G  0 raid0
   └─md0p1   259:0   0    2G  0 part  /raid
sdc          8:32   0    1G  0 disk
└─md0        9:0    0    2G  0 raid0
   └─md0p1   259:0   0    2G  0 part  /raid
sr0         11:0    1   5.1G  0 rom
```

export EDITOR=mcedit

systemctl --force --full edit raid.mount

```
[Unit]
Description=Mount RAID0

[Mount]
What=/dev/md0p1
Where=/raid
Type=ext4
Options=defaults

[Install]
WantedBy=multi-user.target
```

systemctl enable raid.mount --now

mount -a

df -h

```
[root@hq-srv ~]# df -h
Filesystem      Size  Used Avail Use% Mounted on
udevfs          5.0M  100K  5.0M   2% /dev
runfs           2.0G  624K  2.0G   1% /run
/dev/sda2       31G   3.2G   26G  11% /
tmpfs           2.0G   4.0K  2.0G   1% /dev/shm
tmpfs           2.0G    0  2.0G   0% /tmp
tmpfs          392M    0  392M   0% /run/user/0
/dev/md0p1      2.0G   24K  1.9G   1% /raid
```

Задание 3. Настройте сервер сетевой файловой системы (nfs) на HQ-SRV

```
apt-get install nfs-server -y
mkdir -p /raid/nfs
chmod 777 /raid/nfs
mcedit /etc/exports
```

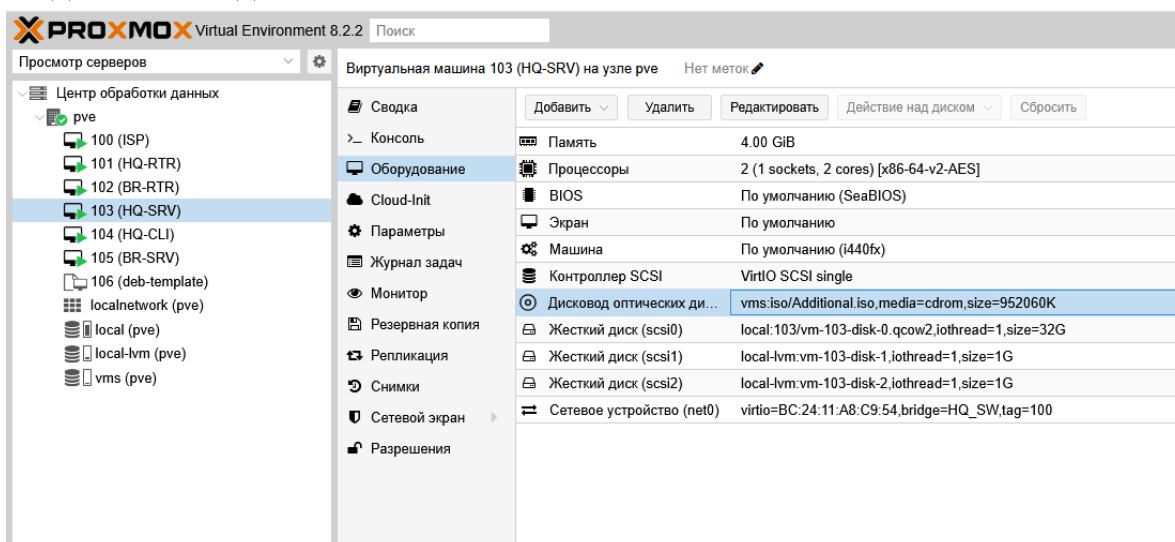
```
exports      [BM--] 50 L:[ 1+ 0  1/ 4] *(50 /
/srv/public -ro,insecure,no_subtree_check,fsid=1 *
/raid/nfs<----->192.168.2.0/28(rw,no_subtree_check)
#/srv/share -rw,insecure,fsid=0,sec=krb5 *
```

```
exportfs -a
systemctl restart nfs-server
systemctl enable --now nfs-server
```

Задание 7. Разверните веб приложение на сервере HQ-SRV

```
apt-get install lamp-server -y
```

Подключение диска Additional.iso:



```
mount /dev/sr0 /mnt/
```

```
ls /mnt/
```

```
(root@hq-srv ~)# ls /mnt
Users.csv  docker  playbook  web
(root@hq-srv ~)#
```

```
cp /mnt/web/index.php /var/www/html
cp /mnt/web/logo.png /var/www/html
mcedit /var/www/html/index.php
```

```
index.php      [-M--] 14
<?php
$servername = "localhost";
$username   = "web";
$password   = "P0ssw0rd";
$dbname     = "webdb";
```

```
mcedit /etc/httpd2/conf/sites-available/default.conf
```



```

default.conf  [-H-]  9 L:1 34+35 69/149] *(2196/5014b) 0060 8x03C
-->#
-->#ServerName www.example.com:80
-->#
--># DocumentRoot: The directory out of which you will serve your
--># documents. By default, all requests are taken from this directory, but
--># symbolic links and aliases may be used to point to other locations.
-->#
DocumentRoot "/var/www/html"
-->#
--># Each directory to which Apache has access can be configured with respect
--># to which services and features are allowed and/or disabled in that
--># directory (and its subdirectories).
-->#
--># First, we configure the "default" to be a very restrictive set of
--># features.
-->#
--><Directory />
--><-----># Summary: Configure for all dirs by default
--><-----># Requires: Directory-root
--><----->Include conf/include/Directory_root_default.conf
--></Directory>
-->#
--># Note that from this point forward you must specifically allow
--># particular features to be enabled - so if something's not working as
--># you might expect, make sure that you have specifically enabled it
--># below.
-->#
-->#
--># This should be changed to whatever you set DocumentRoot to.
-->#
--><Directory "/usr/share/doc/indexhtml/">
--><Directory "/var/www/html/">
--><-----># Summary: Configure for html documents in DocumentRoot
--><-----># Requires: Directory-html
--><----->Include conf/include/Directory_html_default.conf
--></Directory>

```

systemctl enable --now mariadb

mariadb -u root

CREATE DATABASE webdb;

CREATE USER 'web'@'localhost' IDENTIFIED BY 'P@ssw0rd';

GRANT ALL PRIVILEGES ON webdb.* TO 'web'@'localhost' WITH GRANT OPTION;

Exit

mariadb -u web -p -D webdb < /mnt/web/dump.sql

Проверка:

```

[root@hq-srv ~]# mariadb
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 34
Server version: 10.11.14-MariaDB-alt1 (ALT p11)

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

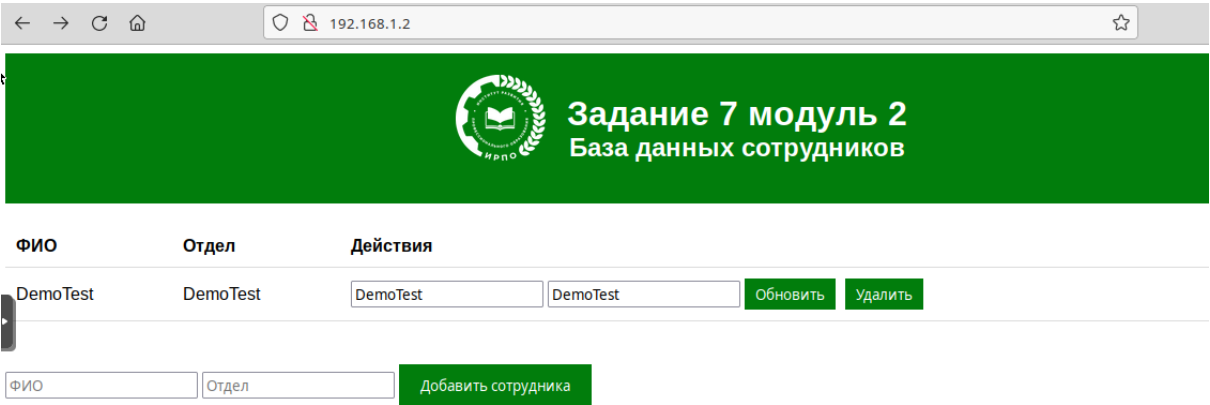
MariaDB [(none)]> USE webdb;
Database changed
MariaDB [webdb]> SHOW TABLES;
+-----+
| Tables_in_webdb |
+-----+
| employees        |
+-----+
1 row in set (0.000 sec)

MariaDB [webdb]> _

```

```
systemctl restart httpd2
systemctl enable --now httpd2
```

Проверка:



BR-SRV(root P@ssw0rd):

Задание 4. Настройте службу сетевого времени на базе сервиса chrony на маршрутизаторе ISP:

```
apt-get update && apt-get install ssh chrony -y  
mcedit /etc/chrony/chrony.conf
```

Комментируем следующие строки:

```
GNU nano 2.7.4                                Файл: /etc/chrony/chrony.conf  
  
# Welcome to the chrony configuration file. See chrony.conf(5) for more  
# information about usable directives.  
#pool 2.debian.pool.ntp.org iburst  
  
# This directive specify the location of the file containing ID/key pairs for  
# NTP authentication.  
keyfile /etc/chrony/chrony.keys  
  
# This directive specify the file into which chronyd will store the rate  
# information.  
driftfile /var/lib/chrony/chrony.drift  
  
# Uncomment the following line to turn logging on.  
#log tracking measurements statistics  
  
# Log files location.  
logdir /var/log/chrony  
  
# Stop bad estimates upsetting machine clock.  
maxupdateskew 100.0  
  
# This directive enables kernel synchronisation (every 11 minutes) of the  
# real-time clock. Note that it can't be used along with the 'rtcfile' directive.  
#rtcsync
```

В конце прописываем:

```
server 172.16.1.1 iburst
```

```
systemctl restart chrony  
systemctl enable chrony
```

Задание 1. Настройте контроллер домена Samba DC на сервере BR-SRV

```
apt-get update && apt-get install task-samba-dc -y  
rm -rf /etc/samba/smb.conf  
samba-tool domain provision  
    Realm: AU-TEAM.IRPO  
    Domain: AU-TEAM  
    Server Role: dc  
    DNS backend: SAMBA_INTERNAL  
    DNS forwarder IP address: 192.168.1.2
```

```
cp /var/lib/samba/private/krb5.conf /etc/krb5.conf
systemctl enable samba --now
```

```
mcedit /etc/samba/smb.conf
```

Добавить:

```
interfaces = lo ens18
```

```
bind interfaces only = yes
```

```
# Global parameters
[global]
<----->dns forwarder = 192.168.1.2
<----->netbios name = BR-SRV
<----->realm = AU-TEAM.IRPO
<----->server role = active directory domain controller
<----->workgroup = AU-TEAM
<----->interfaces = lo ens18
<----->bind interfaces only = yes_

[sysvol]
<----->path = /var/lib/samba/sysvol
<----->read only = No

[netlogon]
<----->path = /var/lib/samba/sysvol/au-team.irpo/scripts
<----->read only = No
```

```
systemctl restart samba
```

```
systemctl enable --now samba
```

Проверка:

```
samba-tool domain info 127.0.0.1
```

```
samba-tool domain info 192.168.3.2
```

```
[root@br-srv ~]# samba-tool domain info 127.0.0.1
Forest           : au-team.irpo
Domain           : au-team.irpo
Netbios domain   : AU-TEAM
DC name          : br-srv.au-team.irpo
DC netbios name  : BR-SRV
Server site      : Default-First-Site-Name
Client site      : Default-First-Site-Name
```

```
samba-tool user create hquser1 P@ssw0rd
```

```
samba-tool user create hquser2 P@ssw0rd
```

```
samba-tool user create hquser3 P@ssw0rd
```

```
samba-tool user create hquser4 P@ssw0rd
```

```
samba-tool user create hquser5 P@ssw0rd
```

```
samba-tool user list
```

```
samba-tool group add hq
```

```
samba-tool group addmembers hq hquser1
```

```
samba-tool group addmembers hq hquser2
```

```
samba-tool group addmembers hq hquser3
```

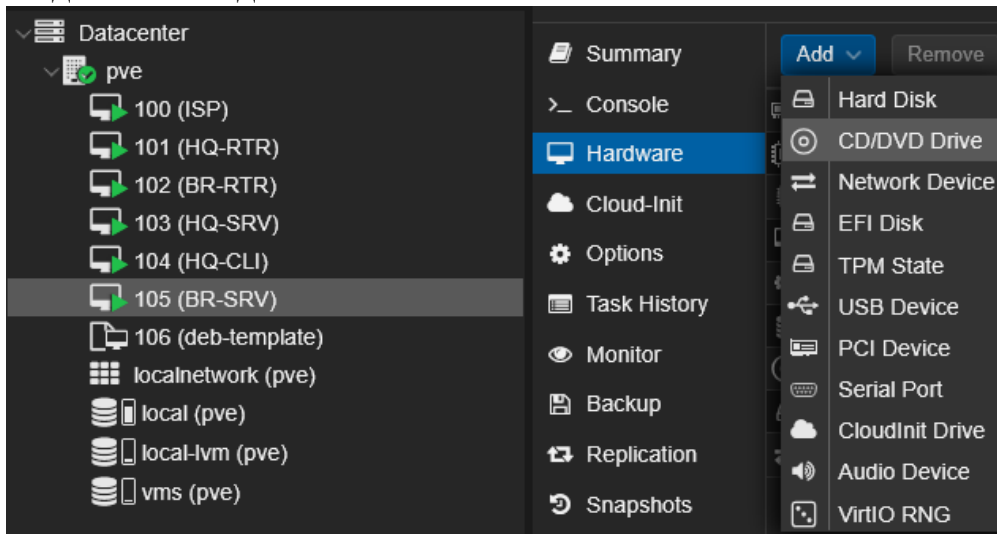
```
samba-tool group addmembers hq hquser4
```

```
samba-tool group addmembers hq hquser5
```

Задание 6. Разверните веб приложение в docker на сервере BR-SRV

```
apt-get install docker-compose docker-engine -y
systemctl enable --now docker
systemctl status docker
```

Подключение диска Additional.iso:



```
mount /dev/sr0 /mnt/
mkdir -p /root/docker/
cp -r /mnt/docker /root/
docker image load -i /root/docker/site_latest.tar
docker image load -i /root/docker/mariadb_latest.tar
mkdir -p /root/testapp/
mcedit /root/testapp/docker-compose.yaml
```

```
docker-compose.yaml  [-M--]  0 L:[ 1+26 27/ 27] *(611 / 611b) <EOF>
services:
  testapp:
    image: site:latest
    container_name: testapp
    restart: always
    depends_on:
      - db
    ports:
      - 8080:8080
    environment:
      DB_TYPE: maria
      DB_HOST: db
      DB_NAME: testdb
      DB_PORT: 3306
      DB_USER: test
      DB_PASS: P@ssw0rd
  db:
    image: mariadb:10.11
    container_name: db
    restart: always
    environment:
      MARIADB_DATABASE: testdb
      MARIADB_USER: test
      MARIADB_PASSWORD: P@ssw0rd
      MARIADB_ROOT_PASSWORD: P@ssw0rd
```

docker compose up -d

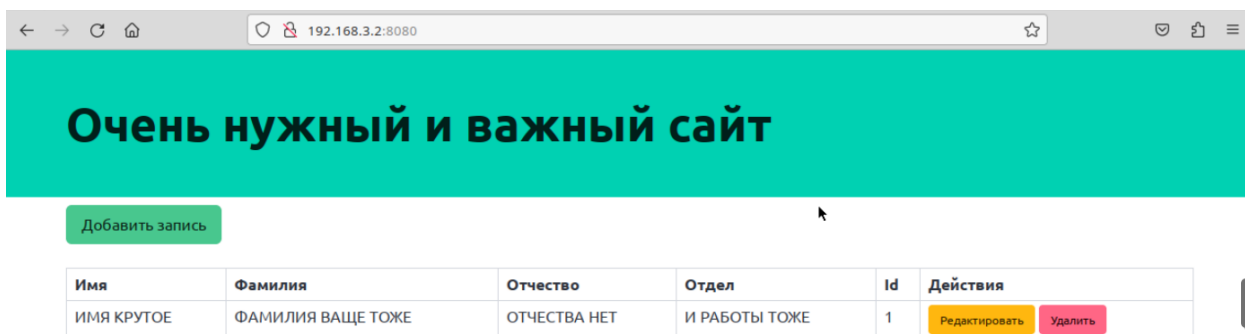
```
[root@br-srv ~]# docker compose up -d
[+] Running 2/2
 ? Container db      Running
 ? Container testapp Started
[root@br-srv ~]#
```

docker ps

```
[root@br-srv ~]# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
8e3e6a7546b5   site:latest    "sh -c 'python3 -m ap"   About a minute ago   Up About a minute   0.0.0.0:8080->8080/tcp, :::8080->8080/tcp   testapp
31d8de0ee451   postgres:15-alpine "docker-entrypoint.sh"   About a minute ago   Up About a minute   5432/tcp                               db
```

Проверка:

На HQ-CLI <http://192.168.3.2:8080>



Задание 5. Сконфигурируйте ansible на сервере BR-SRV

`apt-get install ansible sshpass -y`

`mcedit /etc/ansible/hosts`

```
hosts      [----] 45 L:1 1+ 4 5/ 51 *(346 / 392b) 0117 0x075
[clients]
hq-rtr ansible_host=192.168.1.1 ansible_user=net_admin ansible_port=22 ansible_password=P0ssw0rd
br-rtr ansible_host=192.168.3.1 ansible_user=net_admin ansible_port=22 ansible_password=P0ssw0rd
hq-srv ansible_host=192.168.1.2 ansible_user=sshuser ansible_port=2026 ansible_password=P0ssw0rd
hq-cli ansible_host=192.168.2.2 ansible_user=user ansible_port=22 ansible_password=P0ssw0rd
```

`rm -rf /etc/ansible/ansible.cfg`

`mcedit /etc/ansible/ansible.cfg`

```
ansible.cfg  [-M--] 30 L:1 1+ 3 4/ 41 *(96 / 96b) <EOF>
[defaults]
interpreter_python = python3
host_key_checking = False
inventory = /etc/ansible/hosts
```

Проверка:

ansible all -m ping

```

[root@br-srv ~]# ansible all -m ping
HQ-CLI | UNREACHABLE! => {
  "changed": false,
  "msg": "Failed to connect to the host via ssh: ssh: connect to host 192.168.2.2 port 22: Connection refused",
  "unreachable": true
}
BR-RTR | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
HQ-RTR | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
HQ-SRV | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
[root@br-srv ~]#

```

Ашибочка

На HQ-CLI:

```

apt-get update && apt-get install openssh
systemctl enable --now sshd

```

Возвращаемся на BR-SRV:

```

[root@br-srv ~]# ansible all -m ping
BR-RTR | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
HQ-RTR | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
HQ-SRV | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
HQ-CLI | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
[root@br-srv ~]#

```

HQ-CLI(user P@ssw0rd):

Задание 4. Настройте службу сетевого времени на базе сервиса chrony на маршрутизаторе ISP

```
apt-get update && apt-get install ssh chrony -y  
mcedit /etc/chrony/chrony.conf
```

Комментируем следующие строки:

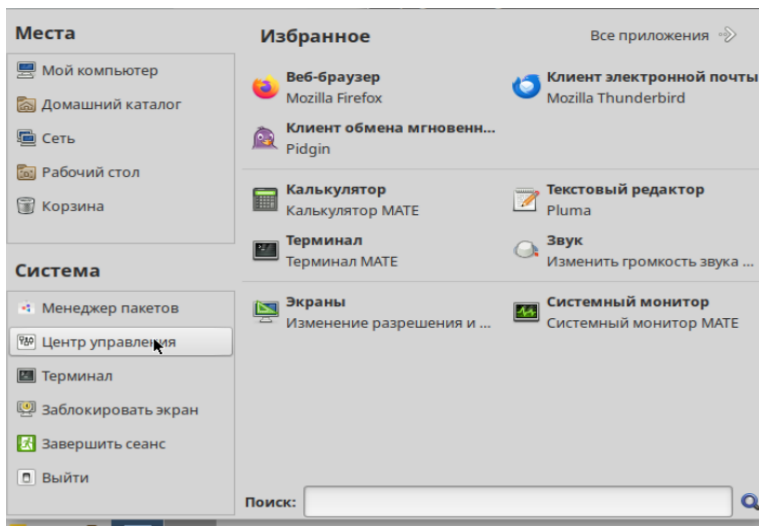
```
GNU nano 2.7.4                                Файл: /etc/chrony/chrony.conf  
# Welcome to the chrony configuration file. See chrony.conf(5) for more  
# information about usable directives.  
#pool 2.debian.pool.ntp.org iburst  
  
# This directive specify the location of the file containing ID/key pairs for  
# NTP authentication.  
keyfile /etc/chrony/chrony.keys  
  
# This directive specify the file into which chronyd will store the rate  
# information.  
driftfile /var/lib/chrony/chrony.drift  
  
# Uncomment the following line to turn logging on.  
#log tracking measurements statistics  
  
# Log files location.  
logdir /var/log/chrony  
  
# Stop bad estimates upsetting machine clock.  
maxupdateskew 100.0  
  
# This directive enables kernel synchronisation (every 11 minutes) of the  
# real-time clock. Note that it can't be used along with the 'rtcfile' directive.  
#rtcsync
```

В конце прописываем:

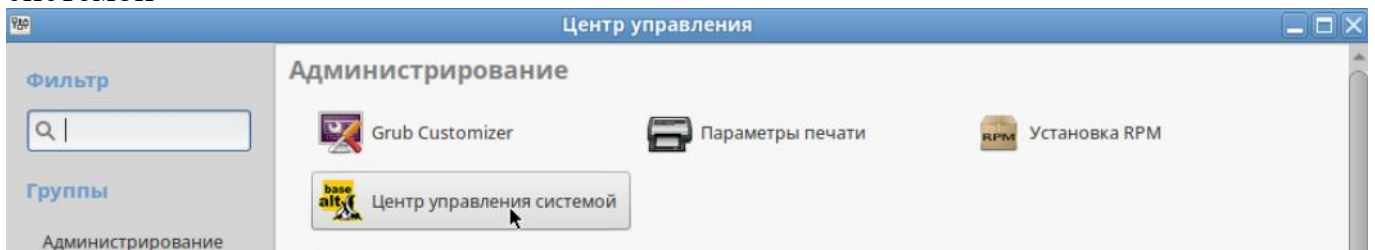
```
server 172.16.1.1 iburst
```

```
systemctl restart chrony  
systemctl enable chrony
```

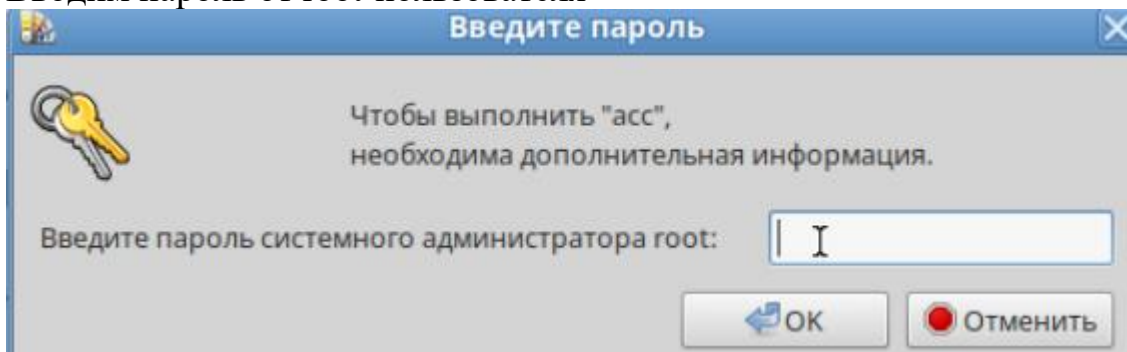
Задание 1. Настройте контроллер домена Samba DC на сервере BR-SRV (вход в домен)



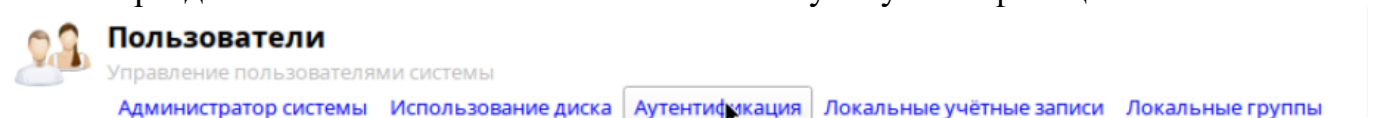
Далее в категории “Администрирование” кликаем по иконке “Центр управления системой”



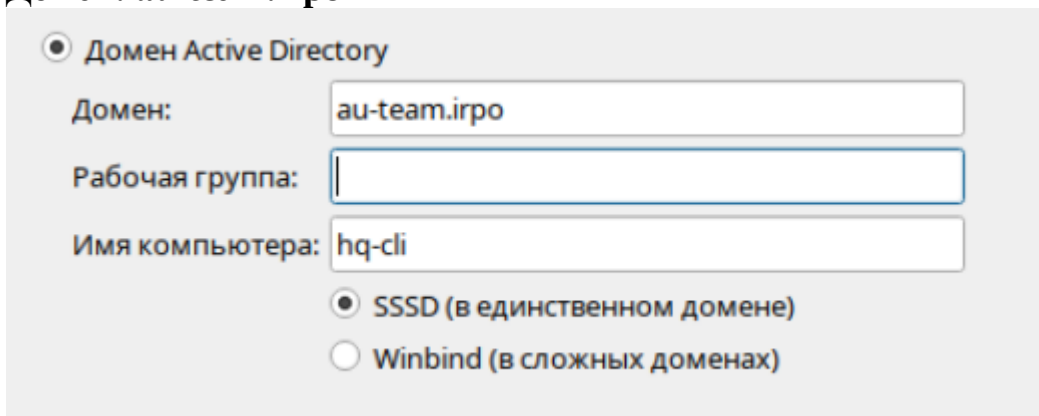
Вводим пароль от root-пользователя



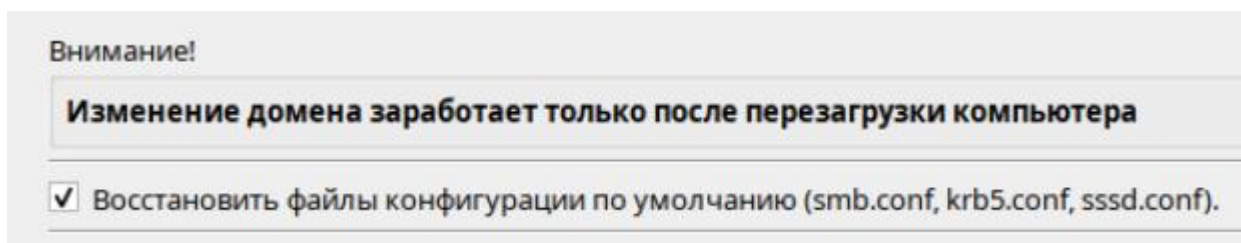
Затем в разделе “Пользователи” кликаем на иконку “Аутентификация”



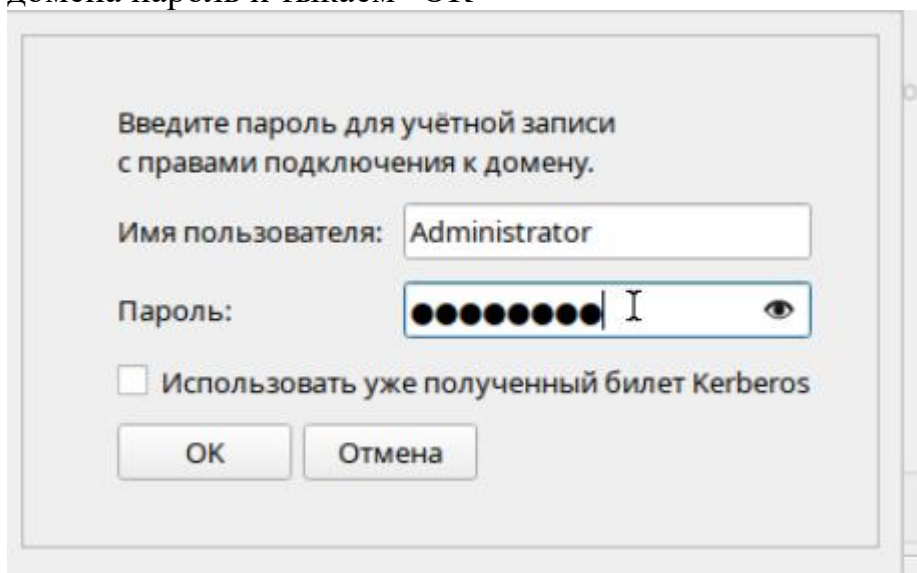
После выбираем пункт “Домен Active Directory” и заполняем поле домена
Домен: au-team.irpo



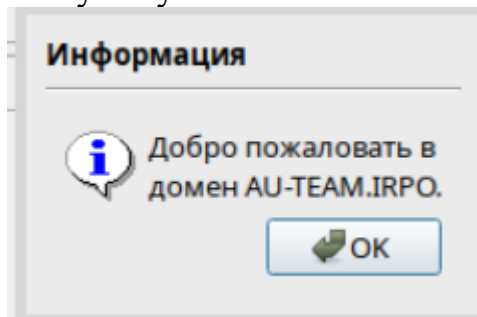
Снизу ставим флажок “Восстановить файлы конфигурации по умолчанию”



Далее в появившемся окне вводим созданный при развертывании контроллера домена пароль и нажимаем “ОК”



В случае успеха появится вот-такое окно!



```
nano /etc/sudoers
```

```
#User_Alias    HQUSERS = hquser1, hquser2, hquser3, hquser4,
%hq            ALL = NOPASSWD: /usr/bin/cat, /bin/grep, /usr/bin/id
#HQUSERS       ALL=(ALL:ALL)  ALL
##
```

```
chmod 4755 /usr/bin/sudo
```

Задание 3. Настройте сервер сетевой файловой системы (nfs) на HQ-SRV

```
apt-get install nfs-clients -y
```

```
mkdir /mnt/nfs
```

```
echo "HQ-SRV:/raid/nfs /mnt/nfs nfs defaults,_netdev 0 0" >> /etc/fstab
```

↑ тут нижнее подчеркивание

```
mount -a
```

```
df -h
```

```
[root@hq-CLI ~]# df -h
Файловая система    Размер  Использовано  Дост  Использовано%  Смонтировано в
udevfs              5,0M      100K        5,0M          2% /dev
runfs              984M      1,3M      983M          1% /run
/dev/sda2           30G       9,0G       20G          32% /
tmpfs              984M          0      984M          0% /dev/shm
tmpfs              984M       12K      984M          1% /tmp
tmpfs             197M       72K     197M          1% /run/user/90201104
tmpfs             197M       76K     197M          1% /run/user/500
tmpfs             197M       88K     197M          1% /run/user/90201107
192.168.1.2:/raid/nfs 2,0G          0       1,9G          0% /mnt/nfs
```

Задание 11. Удобным способом установите приложение Яндекс Браузер на HQ-CLI

`apt-get install yandex-browser-stable -y`

Конечная проверка:

The screenshot shows a web browser window with the address bar displaying 'web.au-team.irpo/'. The page has a green header with a logo and the text 'Задание 7 модуль 2 База данных сотрудников'. Below the header is a table with columns 'ФИО', 'Отдел', and 'Действия'. The table contains one row with the value 'DemoTest' in both the 'ФИО' and 'Отдел' columns. In the 'Действия' column, there are two input fields, both containing 'DemoTest', and two buttons: 'Обновить' (Update) and 'Удалить' (Delete). Below the table is a form with two input fields labeled 'ФИО' and 'Отдел', and a green button labeled 'Добавить сотрудника' (Add employee).

ФИО	Отдел	Действия
DemoTest	DemoTest	DemoTest DemoTest Обновить Удалить

Все студенты ×Задание 7 модуль ×Уведомление ×httpd2.altserve ×Сисадминчик ×Apache2 — ALT ×Проблема при ×+ ▾

← → ↻ 🏠🔒 docker.au-team.irpo ☆🔔📌☰

Очень нужный и важный сайт

Добавить запись

Имя	Фамилия	Отчество	Отдел	Id	Действия
ИМЯ КРУТОЕ	ФАМИЛИЯ ВАЩЕ ТОЖЕ	ОТЧЕСТВА НЕТ	И РАБОТЫ ТОЖЕ	1	РедактироватьУдалить